E-WASTE & ENVIRONMENTAL MANAGEMENT | SEMESTER 5

Toxic Chemicals in E-Waste

Assignment EW-01: Any Five

The five elements covered

Pb	As	Cr	Cd	Hg
Lead	Arsenic	Chromium (VI)	Cadmium	Mercury
Z = 82	Z = 33	Z = 24	Z = 48	Z = 80

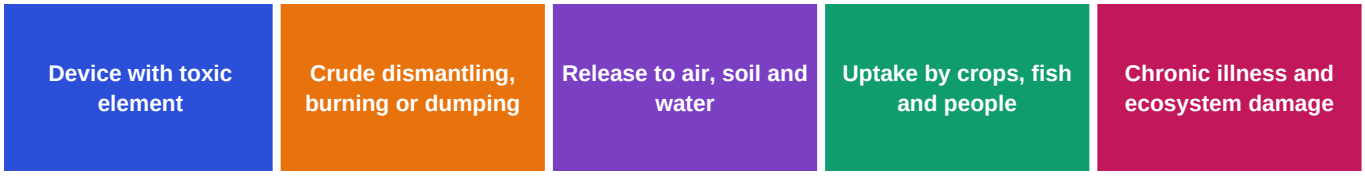
What this report covers

- **Where each toxic element comes from** inside electrical and electronic equipment
- **How it harms people**: key health effects and exposure routes
- **How it harms the environment**: mobility, persistence and bioaccumulation
- **Limits that apply**: RoHS, India's E-Waste (Management) Rules 2022 and BIS 10500 drinking-water limits
- **Side-by-side comparison** and references in IEEE style

1. Why toxic chemicals in e-waste matter

Electrical and electronic equipment contains a complex mix of metals, plastics and additives. When devices reach end of life, those materials become **e-waste**. Globally, about **53.6 million tonnes** of e-waste were generated in 2019, and only **17.4%** was documented as formally collected and recycled [1]. The rest is stored, landfilled, exported or processed informally, often by burning and acid leaching, which turns hazardous materials into direct exposure routes.

Exposure pathway



Regulatory framework

Instrument	What it does
RoHS (EU Directive 2011/65/EU) [2]	Restricts Pb, Hg, Cr(VI), PBB and PBDE to 0.1% by weight and Cd to 0.01% in each homogeneous material
E-Waste (Management) Rules, 2022 [3]	India's rules: extended producer responsibility and reduction of hazardous substances in equipment, following RoHS-type limits
BIS IS 10500:2012 [4]	Indian drinking-water specification with limits for Pb, As, Cr, Cd and Hg
Minamata Convention on Mercury (2013) [5]	Global treaty to phase out mercury products and control emissions

The next five pages profile one element each: **Lead, Arsenic, Chromium (VI), Cadmium and Mercury**.

Pb Lead

The classic neurotoxin of solder and glass

82	0.1%	0.01 mg/L	No safe level
Atomic number	RoHS limit (w/w)	BIS 10500 water limit	for children

Where it is found in e-waste

- Tin-lead solder on printed circuit boards (PCBs)
- Funnel glass of cathode ray tubes (CRTs): often 20-30% lead oxide by weight
- Lead-acid batteries in UPS units and inverters
- Cable sheathing and component terminal finishes

Effects on human health

- Neurotoxic: lowers IQ and impairs attention and development in children
- Anaemia, hypertension and kidney damage in adults
- Crosses the placenta; harms foetal development
- Accumulates in bones and is released slowly over years

Effects on the environment

- Leaches from dumped CRT glass and burned PCBs into soil and groundwater
- Open burning and acid stripping release lead-bearing fume and dust
- Persistent: does not degrade, so contaminated soil stays hazardous

Key point. Informal recycling of CRT monitors and PCBs is a major source of childhood lead exposure around unregulated e-waste sites. [6]

As Arsenic

A metalloid hidden inside semiconductors

33	IARC 1	0.01 mg/L	GaAs
Atomic number	Human carcinogen	BIS 10500 water limit	Key e-waste form

Where it is found in e-waste

- Gallium arsenide (GaAs) in LEDs, laser diodes and high-frequency chips
- Infrared detectors and some solar cells
- Doping agent in silicon semiconductors
- Trace use in older glass and display manufacture

Effects on human health

- Classified by IARC as carcinogenic to humans (skin, lung and bladder cancer)
- Skin lesions, hyperpigmentation and hardening of palms and soles
- Cardiovascular disease, diabetes and neurological effects with chronic exposure
- Acute poisoning: vomiting, abdominal pain, and in high doses, death

Effects on the environment

- Mobilised from discarded chips into groundwater, especially under reducing conditions
- Drinking water is the main exposure route for communities
- Bioaccumulates in rice and some seafood

Key point. Arsenic from GaAs devices is released when chips are crushed, burned or leached with acids during crude recovery of gold and gallium. [7], [10]

Cr Chromium (VI)

Hexavalent chromium: the corrosion coating that corrodes people

24	IARC 1	0.05 mg/L	0.1%
Atomic number	Cr(VI) carcinogen	BIS 10500 (total Cr)	RoHS limit (w/w)

Where it is found in e-waste

- Chromate anti-corrosion coating on steel chassis, screws and fasteners
- Decorative and hard chrome plating on casings
- Data tapes and floppy disks (chromium dioxide)
- Pigments and some plastics

Effects on human health

- Inhaled Cr(VI) causes lung cancer (IARC Group 1)
- Nasal septum ulcers, asthma and chronic bronchitis
- Allergic contact dermatitis and skin ulcers (chrome holes)
- Kidney and liver damage after high exposure

Effects on the environment

- Cr(VI) is highly soluble and mobile in water, unlike the far less toxic Cr(III)
- Contaminates soil and groundwater near dumps and recycling yards
- Toxic to aquatic life even at low concentrations

Key point. Only the hexavalent form is the problem: Cr(VI) is a strong oxidiser that readily crosses cell membranes, whereas Cr(III) is much less toxic. [10], [11]

Cd Cadmium

The slow poison that stays for decades

48	0.01%	0.003 mg/L	10-30 yr
Atomic number	RoHS limit (w/w)	BIS 10500 water limit	Biological half-life

Where it is found in e-waste

- Nickel-cadmium (Ni-Cd) rechargeable batteries
- SMD chip resistors, infrared detectors and semiconductor chips
- Older CRT phosphors and plastic stabilisers
- Contacts, switches and coatings

Effects on human health

- Kidney damage (the critical target organ) and bone demineralisation
- Itai-itai disease in Japan: severe bone pain and fractures from contaminated rice
- IARC Group 1 carcinogen (lung cancer)
- Retained in the body for decades; half-life 10-30 years in the kidney

Effects on the environment

- Highly mobile in acidic soils and readily taken up by crops, especially rice and leafy vegetables
- Enters food chain and bioaccumulates
- Toxic to aquatic organisms

Key point. RoHS applies a ten times stricter threshold to cadmium (0.01%) than to the other restricted metals (0.1%), reflecting its toxicity. [8], [10]

Hg Mercury

Liquid metal, lasting damage

80	0.001 mg/L	0.1%	2013
Atomic number	BIS 10500 water limit	RoHS limit (w/w)	Minamata Convention

Where it is found in e-waste

- Backlight lamps (CCFL) of LCD monitors, laptops and TVs
- Fluorescent tubes and CFLs
- Tilt switches, relays and thermostats
- Button-cell batteries

Effects on human health

- Neurotoxic: tremor, memory loss and developmental damage to the foetus
- Minamata disease from methylmercury in contaminated fish
- Kidney damage and gastrointestinal effects
- Vapour is readily absorbed by the lungs

Effects on the environment

- Evaporates at room temperature and travels globally in the atmosphere
- Bacteria convert it to methylmercury, which biomagnifies up the aquatic food chain
- Persistent in sediments for decades

Key point. Broken lamps and crushed displays release mercury vapour; correct handling and sealed collection of lamps are essential. [9], [5]

7. Side-by-side comparison

Element	Typical e-waste source	Main health effect	Main environmental concern	RoHS limit
Lead (Pb)	Solder, CRT glass, batteries	Neurotoxicity, esp. in children	Soil and groundwater contamination	0.1%
Arsenic (As)	GaAs chips, LEDs, lasers	Skin lesions, cancer (IARC 1)	Groundwater contamination	Not listed
Chromium (VI)	Chromate coatings on steel	Lung cancer, dermatitis	Mobile, toxic to aquatic life	0.1%
Cadmium (Cd)	Ni-Cd batteries, resistors	Kidney and bone damage	Crop uptake, bioaccumulation	0.01%
Mercury (Hg)	LCD backlights, lamps	Neurotoxicity, Minamata disease	Methylmercury biomagnification	0.1%

Common threads

- **Persistence:** none of these elements degrade; they only change chemical form, so contamination lasts for decades.
- **Informal recycling is the main risk:** burning, open acid baths and crushing release the toxic content that intact devices keep contained.
- **Vulnerable groups:** children, pregnant women and informal workers bear the highest exposure.
- **Cadmium, arsenic and Cr(VI)** are IARC Group 1 human carcinogens [10]; **lead and mercury** are above all neurotoxic.
- **Solutions:** design for the environment (RoHS-compliant substitution), extended producer responsibility, and authorised, safe recycling.

8. Conclusion

Lead, arsenic, hexavalent chromium, cadmium and mercury are used in electronics because they solve engineering problems: they make solder flow, semiconductors conduct, steel resist corrosion, batteries recharge and screens light up. The same properties make them hazardous once equipment is discarded. Because these elements persist and bioaccumulate, prevention is far cheaper than cleanup: substitute them in design, collect and recycle devices through authorised channels, and keep informal burning and leaching out of the supply chain.

References

[1] V. Forti, C. P. Balde, R. Kuehr and G. Bel, *The Global E-waste Monitor 2020*. Bonn/Geneva/Rotterdam: UNU, ITU and ISWA, 2020.

[2] European Parliament and Council, "Directive 2011/65/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS)," *Official Journal of the European Union*, 2011.

[3] Ministry of Environment, Forest and Climate Change, Government of India, "E-Waste (Management) Rules, 2022," 2022.

[4] Bureau of Indian Standards, "Drinking Water - Specification," IS 10500:2012, New Delhi, 2012.

[5] United Nations Environment Programme, "Minamata Convention on Mercury," 2013.

[6] World Health Organization, "Lead poisoning," Fact sheet.

[7] World Health Organization, "Arsenic," Fact sheet.

[8] World Health Organization, "Exposure to cadmium: a major public health concern," Geneva, 2019.

[9] World Health Organization, "Mercury and health," Fact sheet.

[10] International Agency for Research on Cancer, "Arsenic, metals, fibres, and dusts," *IARC Monographs on the Evaluation of Carcinogenic Risks to Humans*, vol. 100C, Lyon, 2012.

[11] Agency for Toxic Substances and Disease Registry, "Toxicological Profile for Chromium," U.S. Dept. of Health and Human Services, 2012.